

Clinical Risk Assessment Report: Patient 1

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 36.6% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.65x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM_LGZE.CT), which elevated the patient's absolute risk by +7.3%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.42 [Bottom 2%] (+7.3%)**

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Short Run High Gray-Level Emphasis (CT) (GLRLM_SRHGE.CT) **Value: 1.18 [Top 6%] (+3.9%)**

Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.77 [Avg (75th %ile)] (+2.9%)**

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: -0.07 [Top 15%] (+2.2%)**

Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

High Gray-Level Zone Emphasis (CT) (GLZLM_HGZE.CT) **Value: 1.37 [Top 5%] (+2.0%)**

Highlights the distribution of high-density or highly metabolic active regions clustered within the tumor.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.25 [Bottom 5%] (+1.8%)**

Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Minimum Metabolic Activity (SUVmin) (CONVENTIONAL_SUVbwm) **Value: 0.33 [Avg (72th %ile)] (+0.7%)**

Indicates the areas of lowest metabolic activity within the tumor, which may represent necrotic or poorly perfused regions.

3. Protective / Mitigating Factors (Reducing Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: -0.07 [Avg (45th %ile)] (-3.3%)**

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Machine Learning Feature Attribution (SHAP Decision Path)

